

Auteur: Jonas Saelens
 Studierichting: Informatica
 Oefeningenreeks 1D nr. 23.8

$$z^5 + (3 - 2i)z^4 + 2(1 - 3i)z^3 - 2(1 + 3i)z^2 - (3 + 2i)z - 1 = 0$$

$$\Rightarrow z^5 + (3 - 2i)z^4 + (2 - 6i)z^3 + (-2 - 6i)z^2 + (-3 - 2i)z - 1 = 0$$

$$\begin{array}{c|cccc|c} i & 1 & 3-2i & 2-6i & -2-6i & -3-2i & -1 \\ & & i & 1+3i & 3+3i & 3+i & 1 \\ \hline & 1 & 3-i & 3-3i & 1-3i & -i & 0 \end{array}$$

$$\Rightarrow (z - i)(z^4 + (3 - i)z^3 + (3 - 3i)z^2 + (1 - 3i)z - i) = 0$$

$$\begin{array}{c|cccc|c} -1 & 1 & 3-i & 3-3i & 1-3i & -i \\ & & -1 & -2+i & -1+2i & i \\ \hline & 1 & 2-i & 1-2i & -i & 0 \end{array}$$

$$\Rightarrow (z - i)(z + 1)(z^3 + (2 - i)z^2 + (1 - 2i)z - i) = 0$$

$$\begin{array}{c|ccc|c} -1 & 1 & 2-i & 1-2i & -i \\ & & -1 & -1+i & i \\ \hline & 1 & 1-i & -i & 0 \end{array}$$

$$\Rightarrow (z - i)(z + 1)^2(z^2 + (1 - i)z - i) = 0$$

$$\begin{array}{c|cc|c} -1 & 1 & 1-i & -i \\ & & -1 & i \\ \hline & 1 & -i & 0 \end{array}$$

$$\Rightarrow (z - i)(z + 1)^3(z - i) = 0$$

$$\Rightarrow (z + 1)^3(z - i)^2 = 0$$

$$z_1 = i$$

$$z_2 = -1$$

References